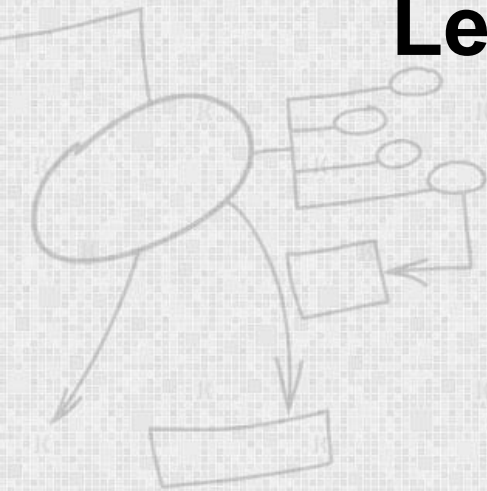


Database Engineering

Lecture 6: Physical Database Design

By

Dr Wasi Haider Butt



Designing Fields

- Field: smallest unit of data in database
- Field design
 - Choosing data type
 - Coding, compression, encryption
 - Controlling data integrity



Choosing Data Types

- CHAR—fixed-length character
- VARCHAR2—variable-length character (memo)
- LONG—large number
- NUMBER—positive/negative number
- INEGER—positive/negative whole number
- DATE—actual date
- BLOB—binary large object (good for graphics, sound clips, etc.)



Example code look-up table (Pine Valley Furniture Company)

PRODUCT File				FINISH Look-up Table	
Product_No	Description	Finish	...	Code	Value
B100	Chair	C		A	Birch
B120	Desk	A		B	Maple
M128	Table	C		C	Oak
T100	Bookcase	B			
...	...	...			

Code saves space, but costs an additional lookup to obtain actual value